LAPTOP REQUEST CATALOG ITEM

Team ID : LTVIP2026TMIDS37778

Team Size : 4

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1. INTRODUCTION

1.1 Project Overview

The ServiceNow Laptop Request Catalog Item Project is developed to automate the process of requesting laptops within an organization. Using the ServiceNow platform, employees can submit laptop requests through the Service Catalog. The system routes the request to the respective manager for approval and then assigns a task to the IT team for fulfillment. The entire lifecycle of the request is tracked within the system, ensuring transparency, accountability, and efficiency.

1.2 Purpose

The purpose of this project is to eliminate manual and email-based laptop request processes. It aims to improve operational efficiency, reduce approval delays, maintain proper documentation, and provide real-time status tracking for both employees and IT administrators.

2. IDEATION PHASE

2.1 Problem Statement

Organizations often handle laptop requests through unstructured methods such as emails or verbal communication. This leads to delayed approvals, lack of proper documentation, difficulty in tracking request status, and inefficiencies in IT asset allocation. A structured and automated system is required to streamline the process.

2.2 Empathy Map Canvas

Employee (Requester)

- Needs a laptop quickly for work tasks.
- Feels frustrated due to unclear request processes.
- Wants transparency in approval and delivery status.

Manager (Approver)

- Needs to validate the request based on necessity and budget.
- Wants a clear view of request details before approval.

IT Team (Fulfillment Team)

- Requires complete and accurate information.
- Needs an organized task management system.
- Wants a system to track request completion.

2.3 Brainstorming

Several solutions were evaluated, including manual registers, spreadsheets, email approvals, and online forms. However, these lacked automation and integration. Implementing a ServiceNow Service Catalog item with workflow automation was selected as the best solution due to its built-in approval mechanisms, task creation capabilities, and reporting features.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

1. Employee logs into ServiceNow.
2. Navigates to the Service Catalog.
3. Selects the Laptop Request item.
4. Fills out the request form with required details.
5. Submits the request.
6. Manager receives approval notification.
7. Manager approves or rejects the request.
8. Upon approval, IT team receives a task.
9. IT allocates and configures the laptop.
10. Request is marked as completed.

3.2 Solution Requirement

Functional Requirements

- Create a Laptop Request catalog item.
- Capture employee details such as name, department, laptop type, and justification.
- Implement manager approval workflow.
- Automatically generate task for IT team.
- Send email notifications at each stage.
- Track request status from submission to closure.

Non-Functional Requirements

- Secure login and role-based access control.
- User-friendly interface.
- High system availability.
- Fast processing time.

3.3 Data Flow Diagram

Level 0 (Context Diagram):
Employee submits request → System processes request → Manager approves → IT fulfills → Employee receives laptop.

Level 1:

- Input: Laptop request form data
- Process: Approval workflow
- Data Store: Request table
- Output: Task generation and status update

3.4 Technology Stack

- Platform: ServiceNow
- Module: Service Catalog
- Workflow Engine: Flow Designer / Workflow Editor
- Database: ServiceNow tables (sc_request, sc_req_item, sc_task)

- Notifications: Email Notification System

4. PROJECT DESIGN

4.1 Problem Solution Fit

The solution directly addresses the challenges of manual laptop request handling. By automating approvals and task assignments, the system ensures faster processing, better tracking, and reduced human errors.

4.2 Proposed Solution

Develop a Service Catalog item named “Laptop Request” that includes form variables, approval workflow, and task automation. The system will notify stakeholders at each stage and maintain records for reporting and auditing.

4.3 Solution Architecture

The architecture consists of:

- User Interface Layer (Service Portal / Service Catalog)
- Application Layer (Workflow / Flow Designer)
- Data Layer (ServiceNow Tables)
- Notification Layer (Email Alerts)

Data flows from user input to approval workflow and finally to IT task assignment.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Phase 1: Requirement Gathering
Phase 2: Catalog Item Creation
Phase 3: Workflow Design
Phase 4: Testing and Debugging
Phase 5: Deployment to Instance
Phase 6: Final Testing and Documentation

The project was completed within the planned timeline with proper testing at each stage.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

- Verified form submission functionality.
- Tested approval workflow routing.
- Ensured task creation for IT team.
- Verified email notifications.
- Tested system response time under multiple submissions.

All test cases passed successfully without major issues.

7. RESULTS

7.1 Output Screenshots

- Service Catalog Homepage
- Laptop Request Form
- Approval Screen
- IT Task Assignment
- Completed Request Status

The system successfully automated the laptop request process.

8. ADVANTAGES & DISADVANTAGES

Advantages

- Reduces manual effort
- Faster approvals
- Real-time tracking
- Centralized request management
- Improved transparency

Disadvantages

- Requires ServiceNow access

- Initial setup complexity
- Dependent on proper workflow configuration

9. CONCLUSION

The ServiceNow Laptop Request Catalog Item Project successfully automates the process of requesting and approving laptops in an organization. It enhances efficiency, ensures transparency, and provides proper tracking of IT asset requests. The project demonstrates how workflow automation improves service management processes.

10. FUTURE SCOPE

- Integration with Asset Management module
- SLA implementation
- Mobile application access
- Automated inventory tracking
- Dashboard and reporting enhancements

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11. APPENDIX

Source Code (if any)

- Catalog Item configuration
- Flow Designer workflow logic
- Notification scripts

Dataset Link

Not applicable (uses internal ServiceNow tables).

GitHub & Project Demo Link

- GitHub Repository Link : https://github.com/kolliMounika9121/laptop-request-catalog-item

- Project Demo Video Link : https://drive.google.com/file/d/1jcET5g27MJMvjHe17bGr2rCj8vOkiwp4/view?usp=drive_link

- Full Project Report : https://drive.google.com/drive/folders/1hzZLmrBsQxzYdUbIhutUPE9MpyhPJGFC?usp=drive_link